

Chapter 19

Using Clip Metadata

DaVinci Resolve has powerful tools for viewing, editing, exporting, and importing metadata associated with each clip in the Media Pool.

Once your metadata house is in order, you can use this metadata in the Cut, Edit, Color, and Fairlight pages to find, sort, and organize the clips in your project, so you can work faster.

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Editing Clip Metadata

Whether you've imported media in preparation for editing, or you've imported a project for grading that resulted in media being imported automatically, once you've added clips to the Media Pool, it would behoove you to consider taking the time to review and add metadata to your clips.

At the very least, it would be valuable for you to use the Metadata Editor that's available in either the Media page or the Edit page to add information to each clip such as a Description, Shot and Scene designations, Take information, and possibly some useful keywords such as Character Names, Shot Framing, Interior or Exterior keywords, and so on. If you're especially ambitious (or you have a very responsible assistant), you could go further and add Shoot Day, Camera Type, Audio Notes, and other valuable information. Much of the metadata that is useful in the day to day work of editing and grading can be found in the Shot & Scene group, but there are many other potentially useful groups as well that you should explore.

Keep in mind that the more metadata you associate with each clip, the more methods you have at your disposal for creating custom Smart Bins (for editing) and Smart Filters (for grading) with which to zero in on the clips you need for any given situation. This will not only make it easier to find what you need, but it'll help you to work faster.

For example, if you've entered enough metadata, then you can create multi-criteria Smart Bins or Smart Filters that let you find the equivalent of "every close-up of Sally inside the diner," or "every long shot of Antonio outside in the parking lot." In a documentary, you could easily isolate "every interview shot of Louis from camera 1," or "every B-roll clip with Robyn." All of this will help you to find media faster for editing, or to quickly isolate similar clips that you need to match together for grading.

Automatically Imported Metadata

In many instances, metadata is also imported along with the media you've added to the Media Pool. For example, media recorded on BMD cameras may have had a variety of metadata entered into the camera or automatically generated by the camera, and this metadata is automatically available in the Metadata Editor. Similarly, Broadcast WAVE files can have quite a bit of metadata entered at the time of recording, such as scene and take numbers and channel names describing each microphone. Still images are imported with EXIF metadata. In all cases, available metadata is imported along with the media and exposed in the Metadata Editor to facilitate workflows where valuable organizational metadata is being entered on set during the shoot or immediately after ingest.

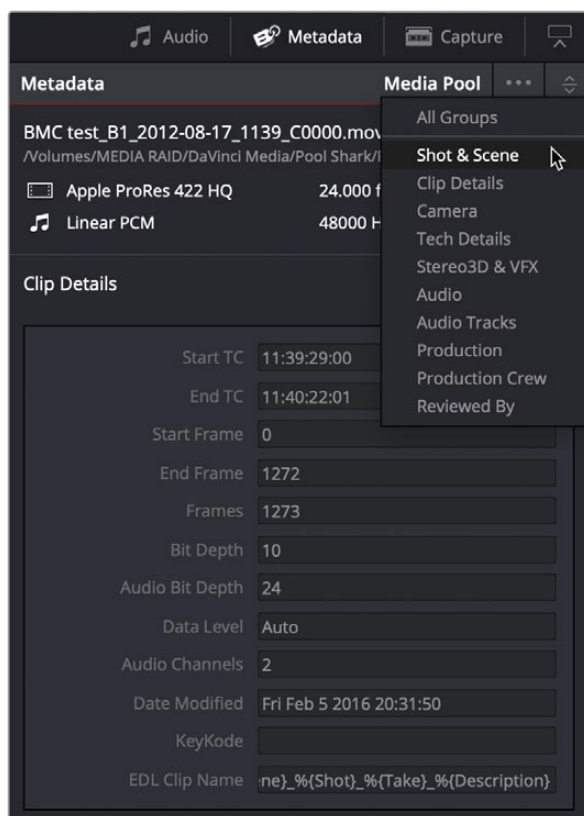
Using the Metadata Editor

Whenever you select a clip in the Media Pool, its editable metadata appears in the appropriately named Metadata Editor (so long as it's displayed). You can use this editor to further massage the metadata of the clips in a project, adding information on set that will be of help later during editing and finishing.

By default, clips initially appear with a set of clip metadata called "Clip Details," that shows some of the most fundamental details of the clip such as start and end timecode, duration, bit depth, and so on.

Because there are so very many metadata fields that are available, two drop-down menus at the top right of the Metadata Editor let you change which set of metadata is displayed.

- **Metadata Presets (to the left):** If you've used the Metadata panel of the User Preferences to create your own custom sets of metadata, you can use this drop-down to choose which one to expose. Surprisingly enough, this is set to "Default" by default.
- **Metadata Groups (to the right):** This drop-down menu lets you switch among the various groups of metadata that are available, grouped for specific tasks or workflows.



Metadata categories drop-down menu

If you want to see a list of every piece of metadata in a clip, you can choose All Groups. Otherwise, you can choose any set of metadata to narrow your focus to just those items of information.

To edit metadata for a single clip:

Select any clip in the Media Pool, and edit whatever metadata fields you require. The edited metadata is immediately saved.

To edit metadata for multiple clips:

- 1 Choose a metadata set using the drop-down menu in the Metadata Editor.
- 2 Select multiple clips in the Media Pool by Shift-clicking, Command-clicking, or dragging a bounding box around them.
- 3 Edit whichever metadata fields you want to. Checkboxes are automatically turned on for any metadata fields you edit.
- 4 When you're done, click the Save button at the bottom of the Metadata Editor. When you've edited metadata for multiple clips at once, you'll be prompted to save your changes if you create a new selection in the Media Pool without clicking the Save button first.

Editing Keywords

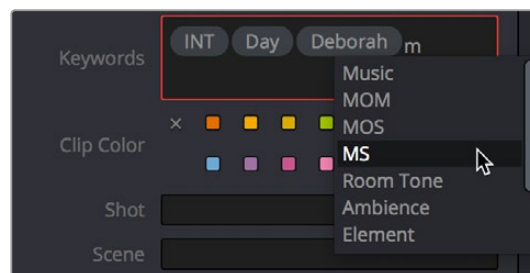
While most metadata in the Metadata Editor is edited via text fields, checkboxes, or multiple button selections (such as Flags and Clip Color), the Keyword field is unique in that it uses a graphical “tag” based method of data entry. The purpose of this is to facilitate consistency with keyword spelling by making it easy to reference both a built-in list of standardized keywords, as well as other keywords that you’ve already entered to other clips.

Once added, keywords are incredibly useful for facilitating searching and sorting in the Media Pool, for creating Smart Bins in the Media and Edit pages, and for use in Smart Filters on the Color page. Reaping these benefits by adding and editing keywords is simple, and works similarly to the method of entering metadata variables.

For more information on metadata variables, see *Chapter 16, “Using Variables and Keywords.”*

To add a keyword:

- 1 Select the Keyword field of the Metadata Editor, and begin typing the keyword you want to use. As you begin typing, a scrolling list appears showing all keywords that are available using the string of characters you’ve just typed.
- 2 To find a specific keyword in the list, start typing that keyword’s name and this list automatically filters itself to show only keywords that contain the characters you’ve just typed. Choose which keyword you want to use in the list using the Up and Down Arrow keys, and press Return to choose that keyword to add.



The keyword list that appears when you type within the Keyword field

As soon as you add one or more keywords, they appear as a graphical tag. To re-edit any keyword, simply click anywhere within the Keyword field to edit it.

To edit a keyword:

- Double-click any keyword to make it editable, then edit it as you would any other piece of text, and press Return to make it a graphical keyword tag again.

To remove a keyword:

- Click any keyword to select it, and press Delete.

TIP: In macOS, any color tags that are set and defined in the Finder can automatically be imported as keywords alongside their media clips. To do so check the “Import Finder tags as Keywords” box in the General Settings panel of the Editing section in the User Preferences.

Editing Metadata Using the File Inspector

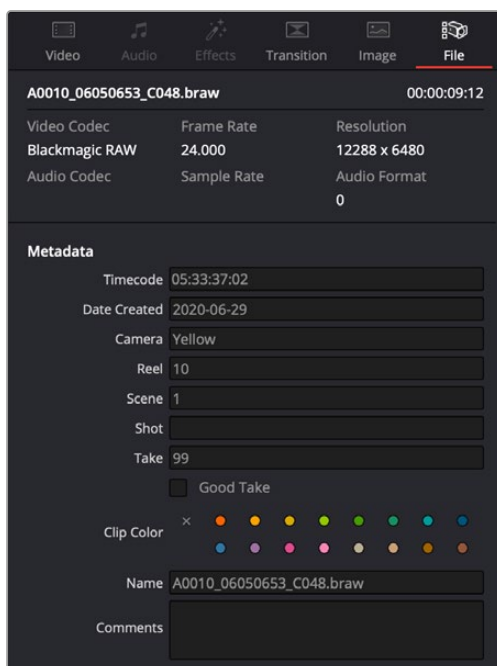
The File tab of the Inspector provides a consolidated way to view and edit a subsection of a clip's most commonly used media file metadata. It's easily accessible in the Inspector across the Media, Cut, Edit, and Fairlight pages. The tab is composed of the following parts:

Clip Details: Presents data about the clip's data format (codec, resolution, frame rate, etc.).

Metadata: Presents a reduced set of common metadata fields for quick user entry.

- **Timecode:** The start timecode of the clip. This field is editable if you want to manually change the clip's starting timecode.
- **Date Created:** The date that the clip was created. This field is editable if you want to manually change the clip's creation date.
- **Camera:** Sets the Camera # metadata.
- **Reel:** Sets the Reel/Card ID.
- **Scene:** The Scene number of the clip.
- **Shot:** The Shot letter/number of the clip.
- **Take:** The Take number of the clip.
- **Good Take:** This checkbox indicates if the clip is a good or circled take.
- **Clip Color:** Assign a specific color to a clip that is reflected in the Timeline.
- **Name:** The clip name field; this can be entered manually.
- **Comments:** Add a text description to the clip.

Auto Select Next Unsorted Clip: When this box is checked, the next clip in the Media Pool is selected when you hit the return button after entering a metadata field, and the cursor is automatically placed in the same field. This allows rapid sequential metadata entry without having to manually click to load each individual clip in the Media Pool. The Next Clip button will select the next clip in the Media Pool, regardless of the checkbox status.



The File Inspector parameters

Tips for Editing Metadata

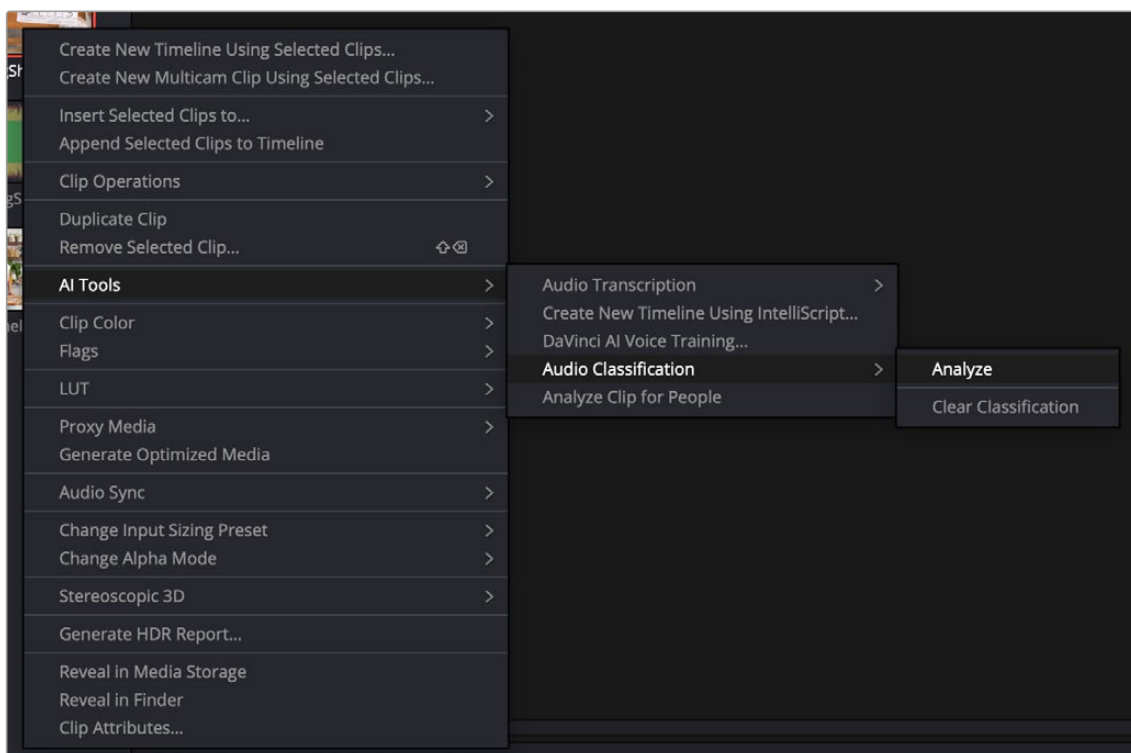
Editing metadata is like taking vitamins. Nobody wants to, but you know you probably should. To encourage you to undertake this task so you can reap the benefits, here are a few pointers.

- Don't start editing until you review your footage and add metadata. If you get into the habit of entering your clip metadata before you get preoccupied with your edit, you'll be in a much better position to edit faster using organizational tools that leverage the metadata you've entered.
- Enter metadata starting with groups of clips and then moving to individual clips. Since the Metadata Editor lets you add metadata for multiple selected clips at once, it becomes easy to select groups of clips based on their thumbnails for entering information such as Scene designations, Interior or Exterior keywords, Character keywords, and Framing keywords. You'll be surprised how fast this goes, and how useful this information is later on, for both editing and grading.
- After you've entered all the metadata you can in groups of clips, then switch to entering clip-specific metadata such as Shot designations, Take numbers, descriptions of action, and other clip-specific keywords.
- There's no right or wrong way to edit or use metadata, but a lack of consistency will make it less useful. For example, if you're identifying each clip that takes place at the same diner, try to use the same keyword or descriptive text. If you call half the shots "diner" and the other half "restaurant," your ability to easily search for all the diner shots will be compromised.

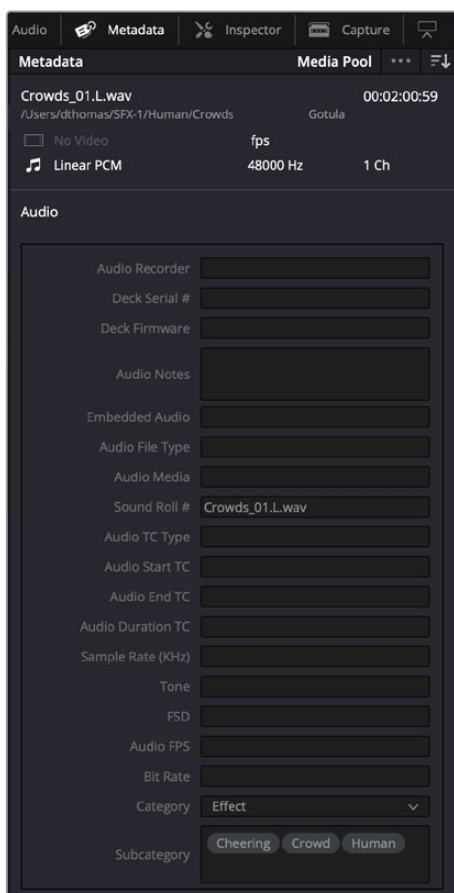
Audio Classification for Clips (Studio Version Only)

You can have DaVinci Resolve's Neural Engine analyze the audio for any clip in the Media Pool, and then automatically assign it a Category and add keywords about its contents in the clip's Subcategory audio metadata. This lets you categorize and organize large amounts of audio clips extremely efficiently by letting the computer do the tedious work of listening to all the media and assigning it metadata for you. Of course, you can then modify or change any metadata necessary in the Metadata Editor in case the computer had miscategorized it.

Each clip analyzed gets Audio Category and Subcategory metadata applied to it. Audio Categories are types of audio commonly used in post-production; as of this writing the available choices are: Dialogue, Effect, Music, Silence, and Uncategorized. Subcategories are more detailed keywords that are assigned based on what sounds the Neural Engine can recognize in the clip. Subcategories can be anything like sirens, water, dog, etc. Audio Category and Subcategory are available as Smart Bin filters that let you quickly organize persistent folders for whatever combination of Categories and Subcategories you desire. For example, you could create a Smart Bin that contains all clips that have both Dialogue (category) and sirens and dogs (subcategories) in them.



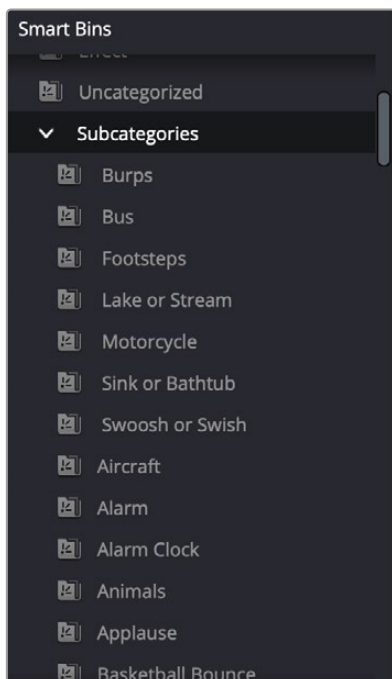
Analyzing an audio clip



The results of the analysis in Category and Subcategory

To Automatically Classify an Audio Clip:

- 1 Select the clip or clips in the Media Pool you want to classify the audio for.
- 2 Right-click on any of the selected clips and select AI Tools > Audio Classification > Analyze from the contextual menu. DaVinci Resolve will work its way through analyzing all the selected clips and will automatically add the metadata to the Audio Metadata for each clip.
- 3 Optionally, you can then review the Audio Metadata for each clip and correct, delete, or add additional metadata manually.



Each Category and Subcategory found by this process automatically creates Smart Bins in the Collections folder, based on that metadata. You can also define your own Smart Bins based on combinations of these keywords.

To Remove Classification Metadata from an Audio Clip:

- 1 Select the clip or clips that have Category and Subcategory metadata that you want to remove.
- 2 Right-click on any of the selected clips and select Audio Classification > Clear Classification from the contextual menu, then click on Remove in the warning dialog box that appears. There is no undo for this feature.

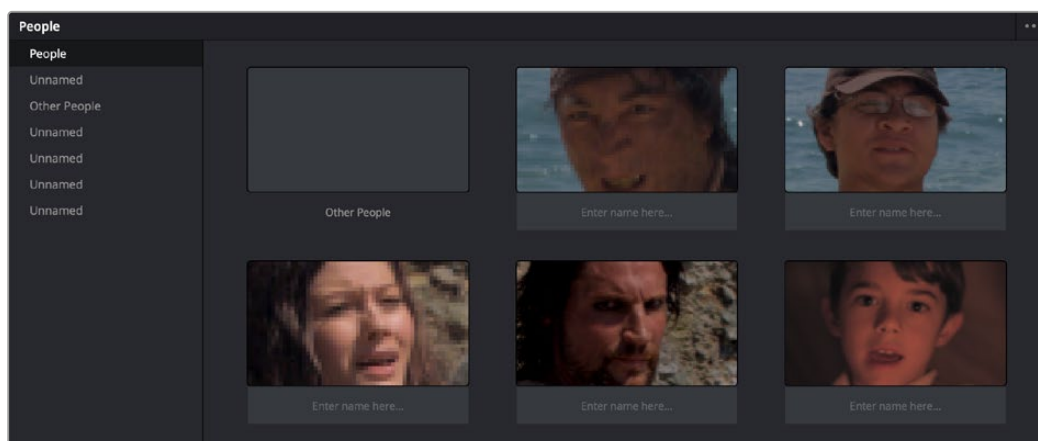
Subcategories folder list showing folders created from analysis

Face Detection to Generate People Keywords

You can select multiple clips in the Media Pool, then right-click the selection and choose AI Tools > Analyze clips for people from the context menu to automatically analyze all selected clips using the DaVinci Neural Engine, identifying faces that can be used to help organize the media. A progress dialog shows you how long until the analysis is finished (you can cancel the operation if necessary).

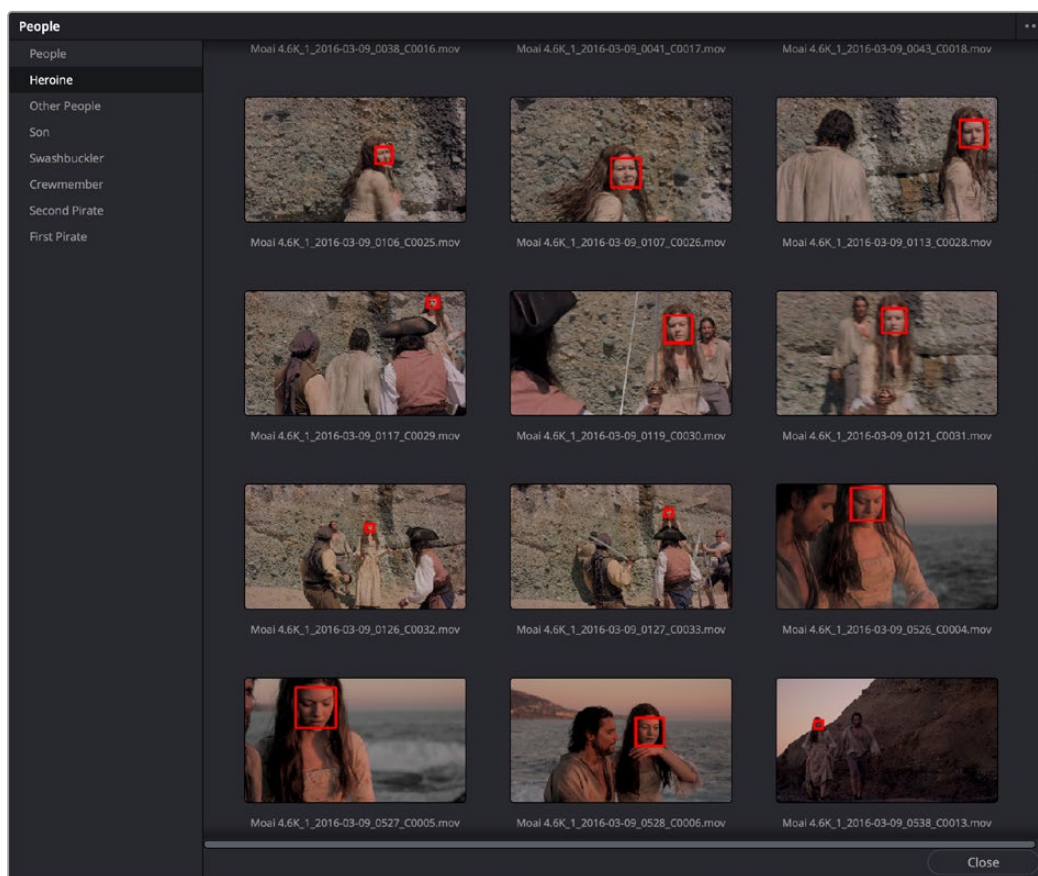
Afterwards, the People Management window appears that shows you the results, automatically organized into a number of bins in a sidebar.

- A “People” bin shows each face that has been recognized as an individual person. Click, pause, then click again underneath any thumbnail to edit the name or role of that person. You must assign a name if you want a keyword to appear for that individual in the People field of the Metadata Editor. Assigning names renames the bins corresponding to each found person and enables retagging to fix mistaken identification.



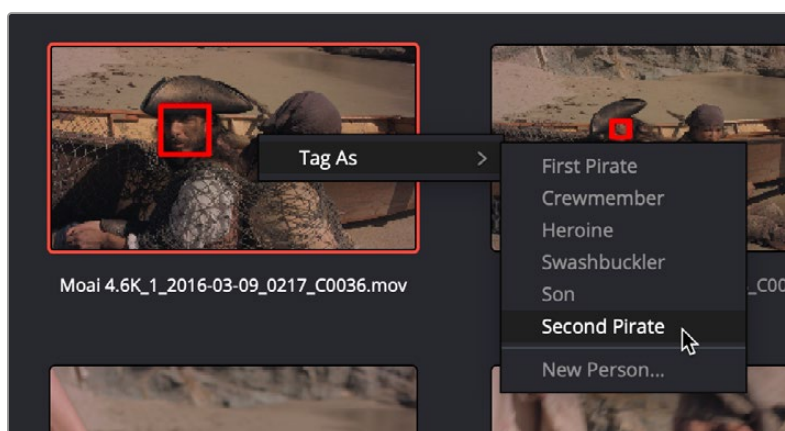
The Face Recognition window seen immediately after a Face Recognition operation

- Individual bins collect all clips with a particular person, allowing you to evaluate whether or not the contents have been identified correctly. If you see an incorrectly identified clip, you can right-click it and re-tag it from the contextual menu, or choose “Untag” if it’s a new person that has not been identified at all.



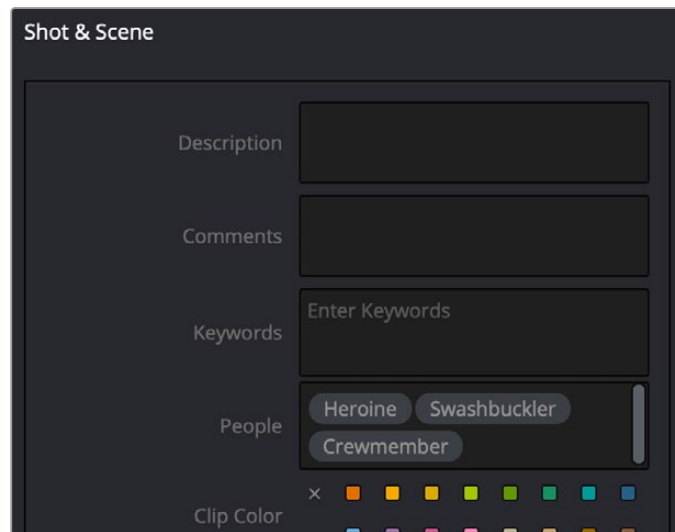
A bin for a particular person lets you evaluate the contents

- An “Other People” bin shows all faces that could not be identified. You can right-click any of these to re-tag it as one of the people that have been already identified, or you can choose New Person if it’s someone who wasn’t initially identified (this sometimes happens when multiple people have very similar features).



The Face Recognition window seen immediately after a Face Recognition operation

Clicking the Close button closes this window and assigns the names you edited as keywords to the People field of the “Shot & Scene” group in the Metadata Editor. Clips with multiple people who have been identified have multiple keywords assigned.



The People keywords field of the Shot & Scene group in the Metadata Editor, populated with who is in that shot

Once People keywords are assigned to one or more clips, a People smart category of smart bins can automatically be created in the Smart Bins sidebar of the Media Pool, making it easy to immediately begin finding clips that have specific people in them. To create this People Smart Bin, select “Automatic Smart Bins for People Metadata” box in the Preferences > User > Editing window.

You can reopen the Face Recognition window at any time to make modifications by choosing Workspace > People. You can reset all faces by clicking the People Management Option menu and choosing “Reset Face Database.”

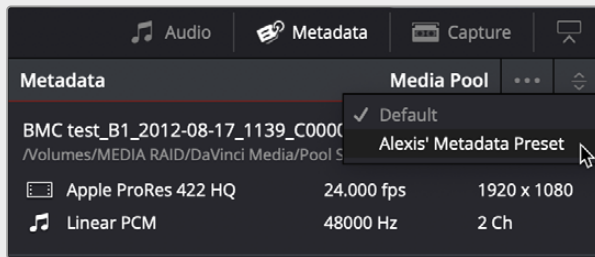
NOTE: A command in the Option menu of the Face Recognition window, Reset Face Database, lets you reset all analyzed results if the results are not acceptable and you don’t want to save the resulting metadata.

Creating Custom Metadata Groups

The Metadata panel in the User Preferences lets you create custom sets of metadata parameters that will be exposed in the Metadata Editor. Using this panel, you can create customized subsets of metadata that are focused on your particular needs.

Presets that you create are available from the Option menu that’s just to the left of the Metadata categories drop-down menu.

Choose any custom preset to restrict the Metadata Editor to only showing the metadata fields in that preset. To see the full set of custom metadata fields you’ve saved to a particular preset, you should set the Metadata Categories drop-down menu to All Groups. To make the full set of metadata fields reappear, just choose default presets in the same drop-down.



Custom Metadata Categories drop-down menu

Making and managing metadata presets is simple.

To create a new metadata preset:

- 1 Open the Metadata panel of the User pane of the Preferences window, and click New.
- 2 Click the checkboxes of every metadata tag you want to include in this preset, or click the checkbox of a group name on the list to include all metadata tags within it.

Every single metadata tag available in DaVinci Resolve appears within one of several groups that appear as a list. To open any group to see its contents, move the pointer over that group's entry on the list, and click the Open button when it appears.

- 3 When you're finished, click the Save button under Metadata Options.
- 4 Click the Save Button for the User Preferences.

To edit an existing metadata preset:

- 1 Select a preset from the list, and click Edit.
- 2 Turn checkboxes on and off to include or exclude whatever tags you need.
- 3 Click the Save button under Metadata Options.
- 4 Click the Save Button for the User Preferences.

To delete a metadata preset:

- Select a preset from the list and click Delete.

Importing and Exporting Media Pool Metadata

Once you've taken the trouble to add metadata to the clips in your project, DaVinci Resolve makes it possible to export metadata from the Media Pool of one project for import into the clips of another project, for instances where you need to move metadata around.

For example, a DIT might have entered a lot of metadata to the DaVinci Resolve project used for generating dailies, but then an impatient editor might have created a separate project to begin editing those dailies. Instead of requiring the editor to enter each clip's metadata all over again, you can export the metadata from the DIT's project and import it into the editor's new project, automatically matching the relevant metadata to each corresponding clip.

To export Media Pool metadata:

- 1 Open a project containing Media Pool metadata you want to export.
- 2 Optionally, select which clips in the Media Pool you want to export metadata for.
- 3 Choose File > Export Metadata From > Media Pool to export metadata from every clip in the Media Pool, or choose File > Export Metadata From > Selected Clips to only export metadata from clips you selected in step 2.
- 4 When the Export Metadata dialog appears, enter a name and choose a location for the file to be written, then click Save. All metadata is exported into a .csv file that can be viewed and/or edited in any spreadsheet application.

If you open the resulting metadata .csv file, the first line is a header that lists what metadata is to be found for each item listed in this document, and in what order. Only metadata fields that have been populated for at least one clip are exported and listed in this header; unused metadata fields in the Metadata Editor or Media Pool are ignored.

This file can now be imported into another project file to reattach the metadata to the same clips.

To import Media Pool metadata:

- 1 Open a project containing clips you want to populate with imported metadata.
- 2 Optionally, select which clips in the Media Pool you want to import metadata to.
- 3 Choose File > Import Metadata To > Media Pool to import metadata to potentially every clip in the Media Pool, or choose File > Import Metadata To > Selected Clips to only import metadata to clips you selected in step 2.
- 4 When the Import Metadata dialog appears, choose a metadata .csv file to import, and click Open.
- 5 When the Metadata Import dialog appears, choose the Import Options you want to use to match the .csv file's metadata to the correct clips in the currently open project. By default, DaVinci Resolve tries to use "Match using filename" and "Match using clip start and end Timecode" to match each line of metadata in the .csv file with a clip in the Media Pool, but there are other options you can use such as ignoring file extensions, using Reel Name, and using source file paths.
- 6 Next, choose which Merge Option you want to use in the Metadata Import dialog.

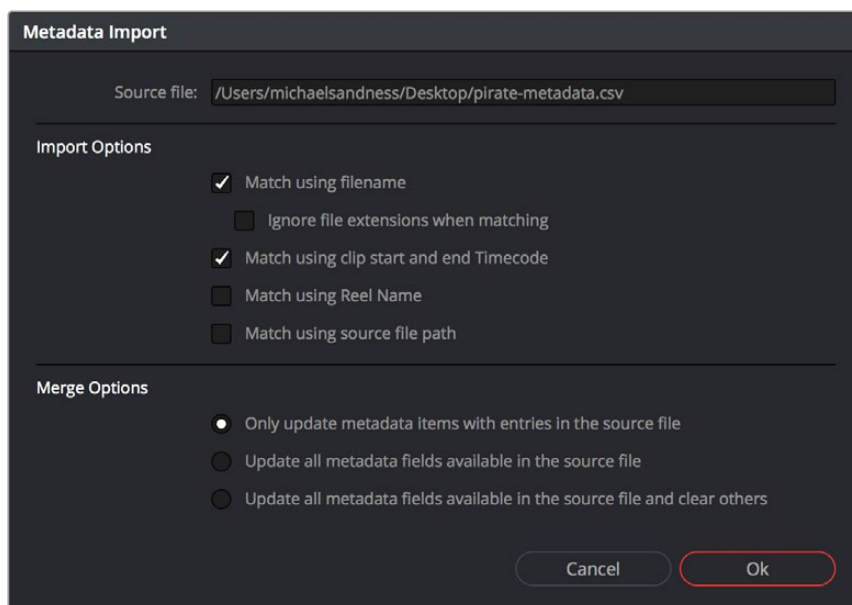
There are three options:

Only update metadata items with entries in the source file: The default setting. Only updates a clip's metadata if there's a valid entry in the imported .csv file. Other clip metadata fields are left as they were before the import.

Update all metadata fields available in the source file: For each clip that corresponds to a line of metadata in the imported .csv file, every single metadata field referenced by the .csv file is overwritten, regardless of whether or not there's a valid entry for that field.

Update all metadata fields available in the source file and clear others: For each clip that corresponds to a line of metadata in the imported .csv file, every single metadata field referenced by the .csv file is overwritten, regardless of whether or not there's a valid entry for that field. Furthermore, metadata fields that aren't referenced by the imported .csv file are cleared of whatever metadata was there before.

- 7 When you're finished choosing options, click Ok and all available metadata from the source .csv file will be imported.



The Metadata Import dialog that lets you choose options for how to match and merge imported metadata

Different Ways of Using Clip Metadata

To encourage you to take advantage of the clip metadata tools that exist in DaVinci Resolve, here's a short list of the many different ways you can use clip metadata to help you work faster.

- Searching for clips in the Media Pool
- Searching for clips in the Timeline
- Sorting the Media Pool by metadata columns in list view
- Creating Smart Bins in the Edit page
- Creating Timeline Filters in the Color page
- Using Metadata to create clip Clip Names
- Displaying Metadata in frame using the Color page Burn In palette

Renaming Clips Using Clip Names

The most fundamental piece of clip metadata is each clip's name, which is used to identify clips nearly everywhere they appear inside DaVinci Resolve. By default, clips show the file name of the corresponding media file on disk. Since the dawn of tapeless recording, however, editors have been stuck with camera original media having names that are not exactly "human readable."

Fortunately, you have the option of entering a more user-friendly clip name to use instead, while preserving the original file name that's critical for maintaining the link between a clip and its media, as well as for tracking an offline clip's corresponding link to the online media from which it originated. There are a few ways you can edit the clip name of a clip.

NOTE: You can also edit the clip names of timelines, compound clips, and multicam clips, so that you can have two sets of naming conventions for these items, one for when you're doing creative editing, and one for when you're doing finishing tasks.

To edit a clip's clip name, do one of the following:

- In the Media Pool's Icon view, click a clip's name once, pause a moment, then click a second time to select the name, type a new name, then press return to accept the name.
- In the Media Pool's List view, the Clip Name mirrors the source clip's file name (hidden by default), but you can click the Clip Name column for any clip to add a new name from scratch.
- With the Clip Name column exposed in the Media Pool's List view, Option-click the Clip Name column for any clip to edit the file name, rather than entering a brand new name.
- To edit the clip name of multiple clips, select all of the clips for which you want to change the clip name, then right-click one of the selected clips and choose Clip Attributes. Open the Name panel of the Clip Attributes window, edit the Clip Name field, and click OK.

After you've changed a clip's clip name, that clip appears in the following places using the clip name instead of the original file name:

- The Media Pool's Thumbnail view
- The name bar of each clip in the Timeline
- The Source Viewer title bar
- The Clip Name field of the Clip Attributes dialog's Name panel

Switching Between File Names and Clip Names

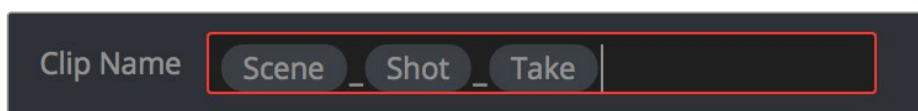
Since different tasks require different information, you have the ability to switch between using clip file names and clip names. For example, finishing editors will probably have more reason to refer to the file name of each clip, making it easier to troubleshoot problems with reconfirming and relinking. Creative editors, on the other hand, will want to use easier-to-read clip names to make it easier to find what they need.

To switch between file names and clip names:

Choose View > Show File Names to toggle between both naming conventions.

Using Metadata to Define Clip Names

If you're an enthusiastic user of clip metadata (and you should be), you can use "metadata variables" that you can add into a field that let you reference other metadata for that clip. For example, you could add the combination of variables and text seen in the following screenshot to define a clip name automatically. Variables, once they've been entered, are represented as graphical tags shown with a background, while regular text characters that you enter appear before and after these tags.



Variables and text characters entered to create a clip name based on a clip's metadata

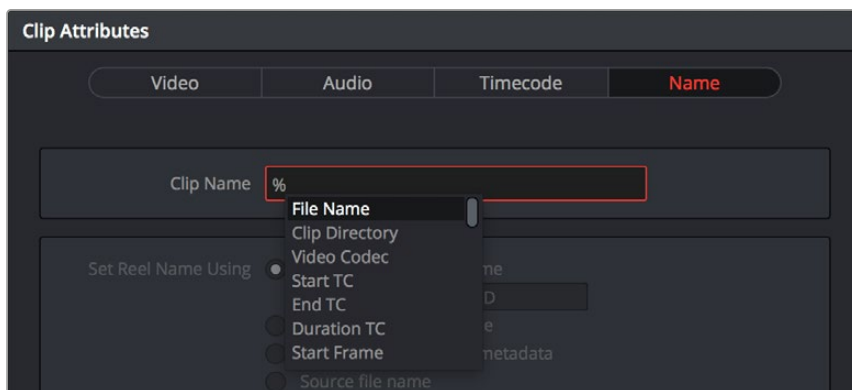
As a result, that clip would display "12_A_3" as its name if scene "12," shot "A," and take "3" were its metadata. When you do this, you can freely mix metadata variables with other characters (the underscore, as in the example above) to help format the metadata to make it even more readable.

Every single item of metadata that's available in the Metadata Editor can be used as a variable, and several other clip and timeline properties such as the version name of a clip's grade, a clip's EDL event number, and that clip's timeline index number can be also referenced via variables.

Since the use of metadata variables is a great way to automatically generate names for multiple clips, you may find it more useful to add metadata variable-driven clip names by selecting all of the clips you want to edit, and opening the Clip Attributes window. By editing the Clip Name field found in the Name panel, you can add a single clip name to all selected clips at once.

To add a variable to a text field that supports the use of variables:

- 1 Type the percentage sign (%) and a scrolling list appears showing all variables that are available.
- 2 To find a specific variable quickly, start typing the characters of that variable's name and this list automatically filters itself to show only variables that contain the characters you've just typed.
- 3 Choose which variable you want to use using the Up and Down Arrow keys, and press Return to choose that variable to add.



The variable list that appears when you type the % character

As soon as you add one or more metadata variables to a clip's Clip Name column and press Return, the string is replaced by its corresponding text. To re-edit the metadata string, simply click within that column, and the metadata variables will reappear. Be aware that, for clips where a referenced metadata field is blank, no characters appear for that corresponding metadata variable in the Clip Name column.

To remove a metadata variable:

Click within a field using variables to begin editing it, click a variable to select it, and press Delete.

For more information on the use of variables, as well as a list of all variables that are available in DaVinci Resolve, see *Chapter 16, "Using Variables and Keywords."*